Graphic Era HILL UNIVERSITY
Established by an Act of the State Legislature of Uttarakhand (Adiniyam Sankhya 12 of 2011)
University under section 2(f) of UGC Act, 1956

# Mid Term (Odd) Semester Examination October 2024

Roll no.....

Name of the Course and semester: B.TECH (ECE), V Semester
Name of the Paper: Object Oriented Programming using C++
Paper Code: TOE 501

Time: 1.5 hour

Maximum Marks: 50

**Note:**

*(i)* Answer all the questions by choosing any one of the sub questions
*(ii)* Each question carries 10 marks.

Q1.

(10 Marks) CO1

a. Create a comparative table highlighting the differences between encapsulation and abstraction, including code snippets for better understanding.

OR

a. What is the concept of inheritance in object-oriented programming? Describe both multilevel inheritance and multiple inheritance, providing examples for each.

Q2.

(10 Marks) CO1

a. What are default arguments in C++? Provide an example to illustrate their use.

OR

b. Describe polymorphism. Explain the difference between compile-time and runtime polymorphism with examples.

Q3.

(10 Marks) CO1/CO2

a. Explain the concept of function overloading in C++.Provide a detailed example demonstrating function overloading, including at least two different functions with the same name but differing parameters

OR

b. What are constructors in C++? Explain the different types of constructors with examples.

Q4.

(10 Marks)CO2

a. What are dynamic constructors in C++? Explain their purpose with a code example, and discuss the advantages of using dynamic constructors.

OR

b. What is operator overloading in C++? Describe how to overload the assignment operator in C++. Provide a code example

Q5.

(10 Marks)CO2

a. What are destructors in C++? Can you explain their key characteristics and provide an example to demonstrate how destructors work?

OR

b. Discuss overloading through friend functions in C++. Provide an example to illustrate your explanation.